

GEM Cluster Timing

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Darklight Software Meeting

June 22, 2026

UPDATE June 30, 2026 – See slide 10

When there is a particle track, when do we detect the clusters in each GEM?

(relative to the trigger time)

- How do we define the time of a cluster?
- Can we distinguish “good” clusters from “junk” clusters based on their timing relative to scintillator hits?

Co-op students have looked into timing of GEM data in simulation.

- Sophie looked at how the ADC value of the “peak” channel of the cluster changes over time samples
- Andy looked at using the GEM as a time-projection chamber to reconstruct track angles

Credit: Sophie Gelinas

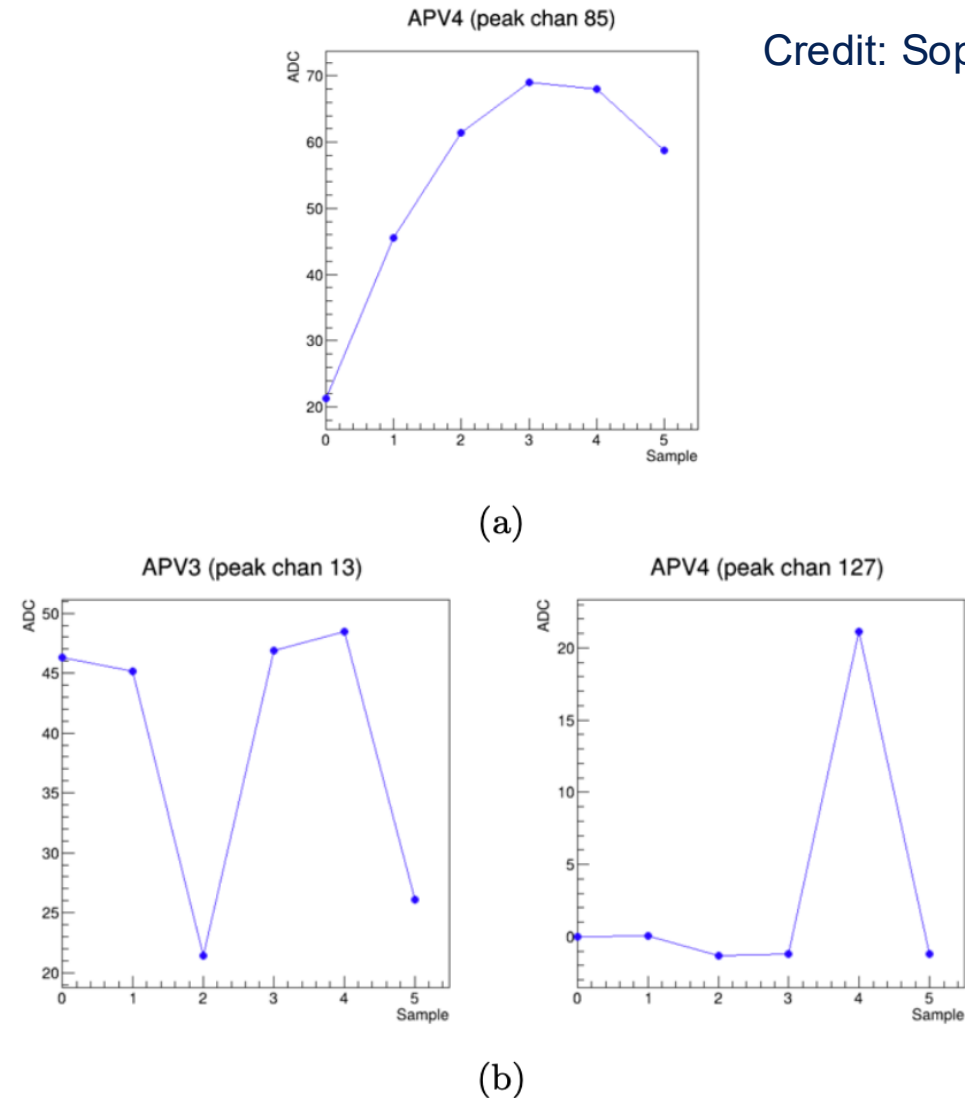


Figure 5: ADC of peak channel vs. sample number for a single APV. (a) shows an x-axis APV, and (b) shows two y-axis APVs.

1. “Max Charge Time”

- Charge $\equiv$ sum of ADC values in all channels in cluster
- How this is calculated:
 - Calculate charge of each sample
 - Calculate “Max Charge Time” as mean sample time weighted by charge of each sample

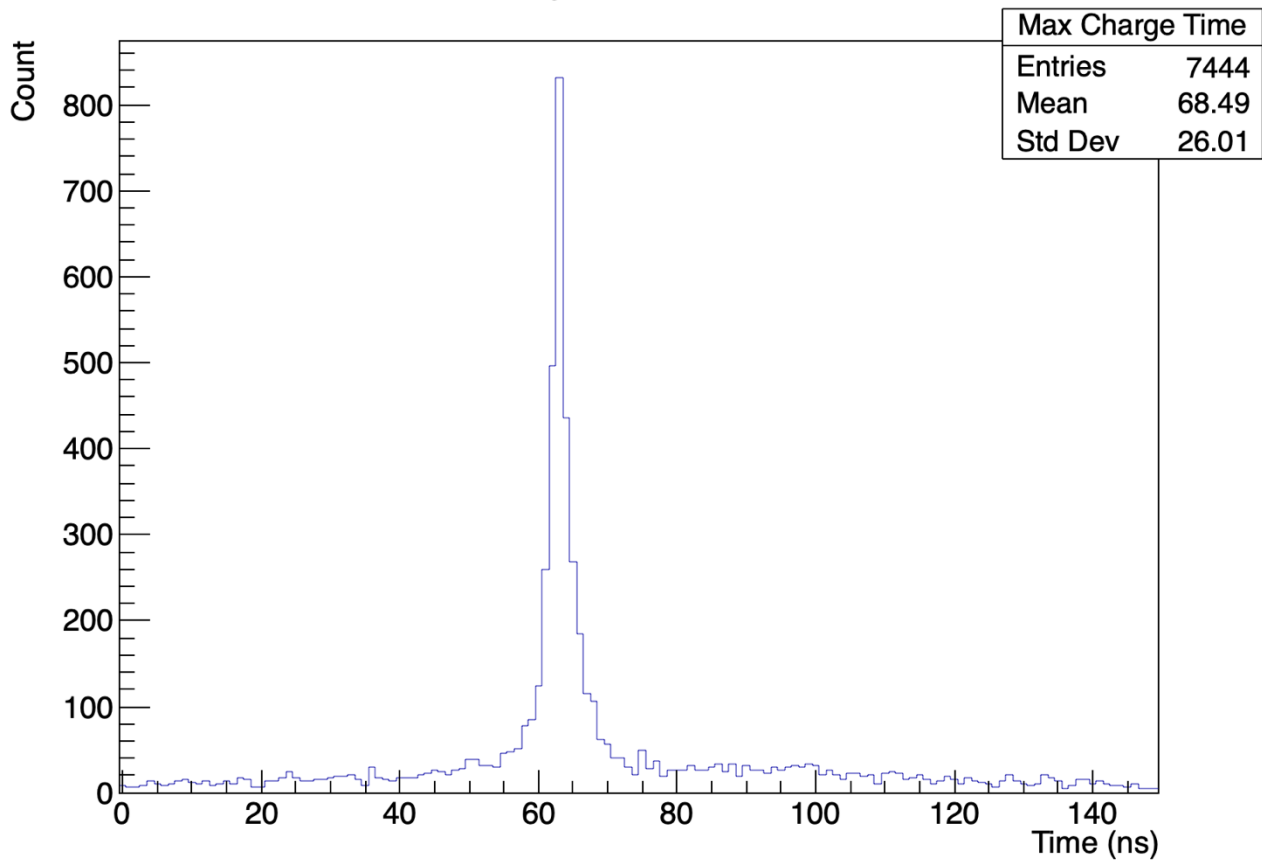
2. “Max ADC Time”

- How this is calculated:
 - Find maximum ADC value in each time sample
 - Calculate “Max ADC Time” as mean sample time weighted by maximum ADC value in each sample

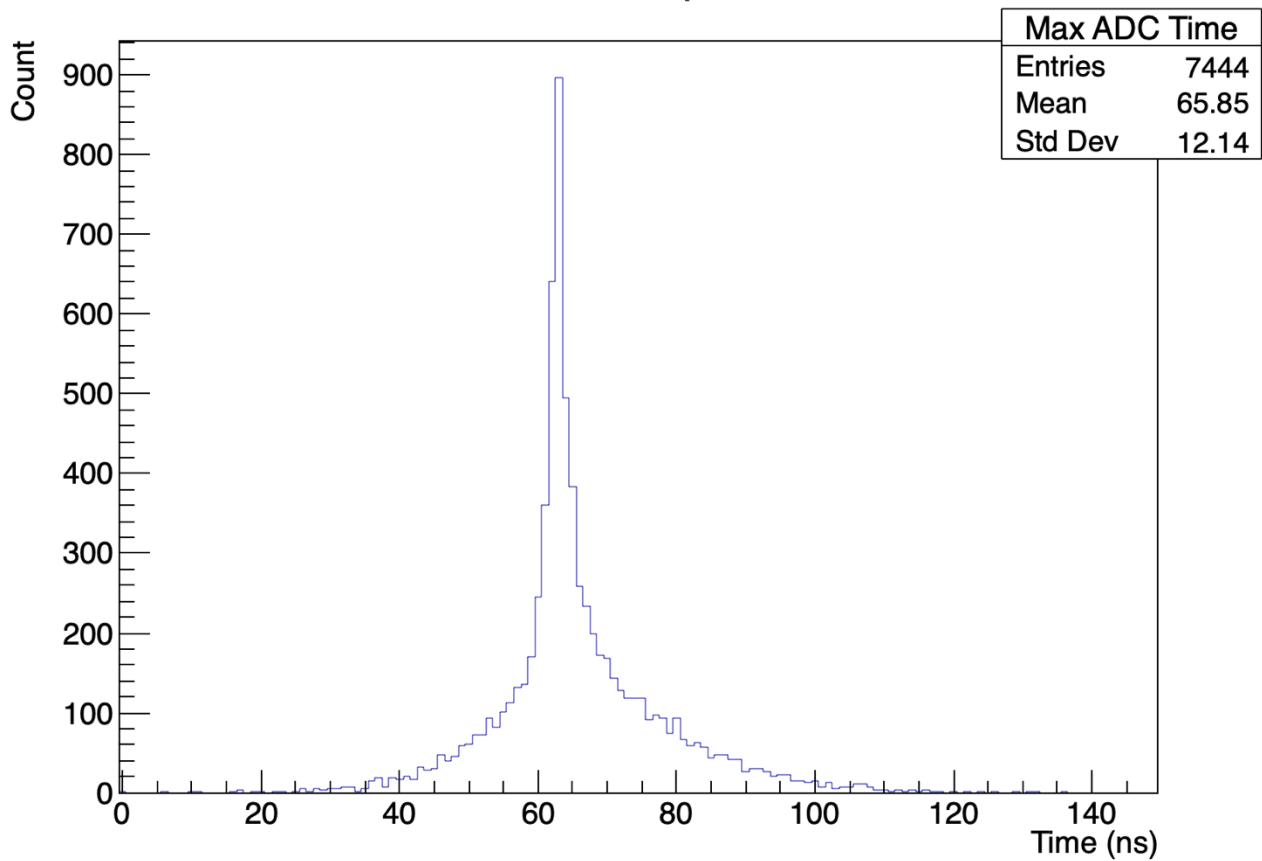
Left Top GEM – X axis cluster times

Run 2679
Cosmic
Full Readout
100k Events

Max Charge Time per Cluster



Max ADC Time per Cluster



- Investigate correlation between first-order cluster time and trigger times in good cosmic ray tracks
- Implement fitting function to improve measure of cluster time

```
242     if (false)
243     { // Previous Method
244
245         input_data data;
246         int index = 0;
247         for (int j = pos - 4; j <= pos + 4; j++)
248         {
249             data.x[index] = j - pos;
250             data.y[index] = values[j];
251             index++;
252         }
253
254         const u_int m = 9, n_par = 3; // Initializing a ton of cminpack stuff
255         int maxfev, mode, nprint, info, nfev, ldfjac, ipvt[n_par];
256         double ftol, xtol, gtol, epsfcn, factor, fnorm, par[n_par], diag[n_par], qtf[
257
258         std::unique_ptr<double[]> fjac = std::make_unique<double[]>(m * n_par);
259         std::unique_ptr<double[]> fvec = std::make_unique<double[]>(m);
260         std::unique_ptr<double[]> wa4 = std::make_unique<double[]>(m);
261
262         par[0] = values[(int)pos];
263         par[1] = 0;
264         par[2] = 0.7;
265
```

Gaussian fitting method



- Instead, the code decides if something is a cluster if its amplitude passes a threshold

```
401     double sig;  
402     double med;  
403     GetMedianAndFilteredStd(pereventhisto, sig, med, 1.0);  
404  
405     if (max - min > 5 * sig)  
406     {  
407         // std::cout << "max: " << max << " min: " << min << " sig: " << sig <<  
408         // std::cout << "max-min" << max-min << " 5*sig: " << 5*sig << std::end  
409
```

- If anyone has information about this code, please let me know

After workfest, my code compiles but crashes when running GEMtrack

- Missing library files
- “terminating due to uncaught exception of type std::bad_alloc: std::bad_alloc”



Thank you

Slide 4

- The code used to generate these histograms had mistakes. Please disregard and do not reference.

Slide 7:

- The information presented here is a misunderstanding of how the code works.